

Yunxiang Guo

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EDUCATION

ETH Zurich

Switzerland, Sep 2023 – Sep 2024

Master Exchange Student

- **Thesis (5.75/6):** Revolutionizing Financial Decision-Making via AI: Reinforcement Learning-Driven Portfolio Optimization and Anomaly Detection in Modern Markets, supervised by [Prof. Dr. Claudio J. Tessone](#) and [Prof. Dr. Josef Teichmann](#).
- **Relevant Modules:** ETHZ: Mathematics for New Technologies in Finance, ETHZ: Optimization for Data Science

University of Zurich

Switzerland, Feb 2021 – Sep 2024

Master of Informatics, Graduated with Magna Cum Laude (5/6)

- **Major:** Artificial Intelligence; **Minor:** Banking and Finance
- **Relevant Modules:** Deep Learning, Blockchains, Reinforcement Learning, Machine Learning, Advanced Topics in Artificial Intelligence, Quantitative Finance

Huazhong University of Science and Technology

China, Dec 2017 - Jul 2020

Bachelor of Management in Business Administration (Double Degree), Achieved GPA: 3.94/4.00

- **Ranked 2nd out of 81 students**
- **Relevant Modules:** Financial Management, Financial Market, Portfolio Analysis and Investment, Finance and Taxation

China University of Geosciences

China, Sep 2016 – Jul 2020

Bachelor of Engineering in Information Engineering, Graduated with Honours (81.3/100)

- **Relevant Modules:** Advanced Mathematics A, Discrete Mathematics, Linear Algebra A, Probability and Statistics A, Algorithm Design and Analysis, Software Project Management

RESEARCH & WORK

World Trade Organization

Switzerland, May 2026 – Now 2026

AI Intern, Integrated Database Unit, Economic Research and Statistics Division

- **AI-powered ETL Pipeline:** develop python-based AI/ML data pipelines with LLM-assisted automation, data preparation, validation, testing, and documentation to support reliable delivery of AI application workflows.
- Contribute to responsible AI solution design by implementing schema intelligence, anomaly detection, prompt/model validation, and human-in-the-loop review for complex multilingual datasets.

The United Nations

Switzerland, Jan 2025 – Dec 2025

Junior Artificial Intelligence Officer (newly established role), International Labour Organization (ILO)

- **AI Project Development:** Serve as the main tech development of advanced AI projects, mainly focus on LLMs Multi agents data search (LangChain, LangGraph); AI Architect (e.g., RAG) based on Labor Market Information Systems (LMIS), SDMX, AI training section, and explore AI cutting-edge technologies; Assist in refining International Standard Classification of Occupations (ISCO) 2028 auto-coding, and identify emerging trends in job market regulations.
- **Supervised** by [Dr. Shutong Ding](#) & [Mr. Greising Edgardo](#), developed an AI handbook for the Department of Statistics, contributed to 113th International Labour Conference, 2025 Generative AI workshop at UNECE, etc.

University of Oxford

UK, Jun 2024 - Dec 2024

Visiting Researcher at Artificial Intelligence and Machine Learning Group

- **Conducted AI research under** [Prof. Dr. Jiarui Gan](#), focusing on multi-agent reinforcement learning systems and game theory, with potential applications in different financial markets.
- **Developed** a robust deep reinforcement learning agent, "dynamic-risk-drl," that uses adaptive risk aversion policies based on CVaR to handle volatility in financial markets.
- **Enhanced abilities** in interdisciplinary problem-solving by applying AI and game theory principles to real-world financial systems and in improving the understanding of risk mitigation in volatile markets.

Pamalican Asset Management Limited (500M USD New Fund)

China (Hong Kong), Jan 2024 - Jun 2024

AI Researcher in Portfolio Management

- **Developed** an AI-driven trading strategy using risk-averse reinforcement learning to optimize trading decisions across major Asian stock markets, including Japan, Hong Kong, Australia, and Singapore. This approach emphasized balancing risk and return, adapting to the unique dynamics and volatility of each market.
- **Collaborated** with and mentored by [Mr. Jamie Goodman](#), CEO of the fund, to refine the strategy, leveraging his expertise to align the model with real-world trading objectives and risk management practices.
- **Achieved 40% to 60%** annual returns in backtesting while maintaining a portfolio risk level below **10%**, showcasing the strategy's profitability and risk management.

ETH Zurich AI Center

Switzerland, Feb 2023 – Jun 2023

Research Team Lead (Team of 4) | AI Researcher

- **Combined** numerical rewards with human preferences to iteratively refine the reward model, aligning the agent's behavior more closely with intended human goals and enhancing adaptability in reinforcement learning tasks, supervised by [Prof. Dr. Mennatallah El-Assady](#) and [Prof. Dr. Giorgia Ramponi](#).
- **Demonstrated** that adding human preference feedback improves the performance of standard RL algorithms (such as SAC and PPO), with experiments showing that diverse state representations can further enhance alignment with human-desired behaviors.
- **Presented a poster at the AI Center Projects in Machine Learning Research 2023.**

UZH Blockchain Center

Switzerland, Feb 2022 - Feb 2023

Project Lead (Team of 2) & Researcher (Risk Modeling / Deep RL)

- **Led** an end-to-end project building a risk-aware deep RL framework for statistical arbitrage, integrating time-consistent dynamic risk measures (DCVaR/CVaR) into value estimation and training.
- **Implemented** a transaction-level actor-critic pipeline with soft-updated target networks and bounded continuous actions; validated on OU stochastic processes, demonstrating stable convergence and positive cumulative wealth.
- **Project grade: 5/6**

China National Engineering Research Center for Geographic Information System

China, Sep 2017 - Jul 2020

Research Project Lead (Team of 2)

- **Led the College Students' Innovative Entrepreneurial Training Plan Program** under [Prof. Dr. Yu Wenhao](#), evaluating the long-term economic impact of the Beijing-Guangzhou High-Speed Railway (BGHSR) on cities along the rail line using night-time light (NTL) data.
- **Revealed** that the BGHSR significantly boosted economic growth in first-tier and new-first-tier cities but negatively impacted some second- and third-tier cities, reshaping the spatial economic structure along the corridor.
- **Published a paper in Socio-Economic Planning Sciences.**

PUBLICATIONS & WORKING PAPERS

[1] **Guo, Y.** (2025). Shaping the United Nations' future with responsible AI and inclusive analytics. [United Nations newSpecial Magazine](#).

[2] Ding S., **Guo, Y.**, Greising, E., Lei, W., Pogor, I., & Vittorelli, C (2025). Labour Market Intelligence: Drafting analysis reports with generative AI. [Generative AI and Official Statistics Workshop 2025](#), *United Nations Economic Commission for Europe*.

[3] **Guo, Y.**, Yu, W., Chen, Z., & Zou, R. (2020). Impact of high-speed rail on urban economic development: an observation from the beijing-guangzhou line based on night-time light images. *Socio-Economic Planning Sciences*, [100905](#).

ACADEMIC ACTIVITIES

Shibaura Institute of Technology

Japan, Dec 2019

Participant of Cross-cultural Engineering Project (CEP)

- Resort revenue optimisation for Nasu and visual programming for KODENSHA intelligent automation company.
- **Increased tourist volume by 11%** over one year and **reduced** Nasu's budget by **approximately 300K per year** by providing innovative visual programming solutions and strategic guidance, significantly lowering company costs associated with hiring external experts.

University of Cambridge & Oxford

England, Jul 2019 - Aug 2019

Summer Institute

- **Relevant Modules:** Strategic Management and Investment Analysis, Artificial Intelligence and Numerical Computing.

EXTRACURRICULAR ACTIVITIES

The United Nations

Switzerland, Jan 2025 – Jun 2025

Co-President, Intern Board (54 interns), at International Labour Organization

- **Leadership & Coordination:** Oversees the Intern Board, ensuring the successful execution of responsibilities, resolutions, and activities, while scheduling and leading meetings to align team efforts.
- **Strategic Partnership & Representation:** Serves as the primary representative in communications with HRD, the Staff Union, and other key organizations such as WTO, ITU, etc, ensuring strong collaboration and effective execution of board initiatives.

BCOLN Challenge Task Award 2023

Switzerland, Jun 2023

Winner

- **Best Distributed Application (dApp) Implementation:** Achieved 1st place (out of 36) for the "Lottery dApp," a cryptocurrency roulette application on the blockchain.
- **Innovation in Cryptocurrency:** Developed a pioneering cryptocurrency application within a new financial systems blockchain, enhancing user engagement and platform functionality.

Google Marketing Live 2022

Switzerland, May 2022

Ambassador for Privacy & Measurement

- **External Tech Support:** Developed neural network prediction models and advertising delivery strategies focused on privacy, utilizing personal data without cookies.
- **Client Retention:** Successfully maintained relationships with high-profile clients: BMW, UBS, and HSBC.

SKILLS

Languages: Chinese - Native, English - Professional, French - Intermediate, German - Intermediate.

Skills: C++, Python, R, C#, VS code, SQL, PyCharm, PyTorch, LaTeX, Docker, Git.

Interests: Sports